Robustness Analysis of PointPillars Under LiDAR Sparsification in 3D Object Detection

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Abstract

3D object detection is crucial for autonomous vehicle perception, yet high-resolution 64-beam LiDAR sensors impose significant monetary and computational costs. This paper evaluates the operational robustness of the PointPillars architecture subjected to two controlled data sparsification regimes without model retraining: uniform random point drop ($\rho \in \{1.0, 0.75, 0.50, 0.25\}$) and vertical beam reduction (64 to 32 to 16 rings). Evaluating performance on the KITTI validation dataset via mAP_{40} metrics across standard difficulties (Easy, Moderate, Hard) and range-segmented spatial intervals (0–20 m, 20–40 m, 40–70 m), we quantify spatial degradation patterns. Our findings demonstrate that PointPillars exhibits strong resilience under uniform density reduction down to 50%, whereas vertical beam reduction induces severe performance decay, particularly for distant targets (> 40 m) and small object categories.

1 Introduction

3D perception is fundamental for autonomous driving, providing precise geometric structure through LiDAR point clouds. However, standard 64-beam mechanical LiDAR sensors (e.g., Velodyne HDL-64E) are constrained by high cost, mechanical complexity, and computational overhead. Autonomous vehicles frequently operate using lower-beam hardware (e.g., 32 or 16 rings) or encounter adverse environmental conditions that cause significant point loss.

Understanding how pre-trained 3D detectors degrade under hardware reduction is essential for sensor deployment and fail-safe perception design. In this work, we evaluate an off-the-shelf PointPillars model under synthetic sparsification on the KITTI benchmark.

1.1 Main Contributions

- **Sparsification Pipeline:** An automated preprocessing framework supporting uniform random point drop and realistic elevation-based vertical ring subsampling.
- **Range-Aware Evaluation Framework:** A depth-aware evaluation module segmenting the camera Z-axis into *Near*, *Medium*, and *Far* spatial zones.
- **Quantitative Degradation Analysis:** A comparative benchmark assessing class-wise sensitivity and depth breakdown under hardware sparsification.

2 Methodology

2.1 PointPillars Architecture Overview

PointPillars converts unstructured LiDAR point clouds $P = \{p_i = (x_i, y_i, z_i, r_i)\}_{i=1}^{N}$ into 2D pseudo-images to utilize efficient 2D convolutional networks. The architecture consists of three principal modules:

1. **Pillar Feature Encoder:** Point clouds are partitioned into vertical pillars along the x-y plane. A simplified PointNet extracts a feature vector per pillar, constructing a sparse pseudo-image tensor.
2. **2D CNN Backbone:** A feature extraction network utilizing top-down convolutional layers and feature pyramid networks (FPN) to aggregate multi-scale contextual features.
3. **Single-Stage Detection Head:** A 2D SSD head predicting 3D bounding box parameters $(x, y, z, h, w, l, \theta)$ and class probabilities using 3D anchors.

2.2 LiDAR Subsampling Pipeline

To evaluate degradation without modifying network weights, point clouds are processed through two sparsification operations prior to inference:

2.2.1 Uniform Random Point Drop

To simulate point density loss caused by atmospheric absorption or reduced sensor sensitivity, input point clouds are sampled at fixed keep ratios $\rho \in \{1.0, 0.75, 0.50, 0.25\}$. Given an initial point cloud of size N, $N_{\text{keep}} = \lfloor \rho \cdot N \rfloor$ points are selected uniformly at random across the spatial domain without replacement.

2.2.2 Vertical Beam Reduction

To simulate hardware downgrades from 64-ring sensors to 32-ring or 16-ring sensors, points are filtered based on their vertical elevation angle θ. For a point $p_i = (x_i, y_i, z_i)$, elevation is computed as:

$$\theta = \arcsin\left(\frac{z_i}{\sqrt{x_i^2 + y_i^2 + z_i^2}}\right) \tag{1}$$

The dynamic elevation range $[\theta_{\text{min}}, \theta_{\text{max}}]$ is divided into $N_{\text{orig}} = 64$ ring channels. Downsampling strides $S = N_{\text{orig}}/N_{\text{target}}$ ($S = 2$ for 32 rings, $S = 4$ for 16 rings) filter points belonging to non-selected ring channels.

2.3 Spatial Depth Range Evaluation Framework

Standard KITTI mAP_{40} metrics integrate predictions over the full range (0–70 m), obscuring depth-specific failure modes. We partition ground-truth boxes and model predictions along the camera Z-axis into three distinct depth ranges:

- **Near Range:** $Z \in [0, 20)$ m (high point density, immediate collision zone).
- **Medium Range:** $Z \in [20, 40)$ m (standard operational driving range).
- **Far Range:** $Z \in [40, 70]$ m (sparse point returns, long-range perception).

3 Experimental Results & Discussion

3.1 Quantitative & Class Sensitivity Analysis

We evaluate the PointPillars baseline on the KITTI validation set across all subsampling regimes using 3D mAP_{40} at standard IoU thresholds (0.70 for Car, 0.50 for Pedestrian and Cyclist). Table 1 summarizes the results across difficulty levels.

Table 1: 3D mAP_{40} performance across sparsification regimes on KITTI Validation Set.

Subsampling Regime	Car (IoU 0.70)			Pedestrian (IoU 0.50)			Cyclist (IoU 0.50)		
	Easy	Mod.	Hard	Easy	Mod.	Hard	Easy	Mod.	Hard
Baseline (100%, 64 rings)	67.40	62.71	59.77	53.82	49.63	44.13	48.11	43.70	42.59
Random Drop ($\rho = 0.75$)	65.34	59.62	56.88	55.25	49.38	44.93	50.24	40.88	40.34
Random Drop ($\rho = 0.50$)	65.17	54.77	51.99	54.77	49.56	44.68	46.91	35.78	35.57
Random Drop ($\rho = 0.25$)	55.26	43.81	39.42	45.55	40.10	35.57	28.50	20.25	21.25
Beam Reduc. (32 rings)	68.04	56.61	52.01	50.47	45.04	40.38	41.82	29.62	30.63
Beam Reduc. (16 rings)	49.00	36.80	33.85	31.75	26.97	24.00	18.90	13.45	14.93

Key Observations: 1) PointPillars demonstrates high robustness under uniform random drop down to $\rho = 0.50$. Local PointNet feature extraction inside pillars aggregates sparse point features without loss of geometric cues. At $\rho = 0.25$, performance drops at 43.81% in Moderate Car mAP_{40}.

2) Vertical beam reduction degrades performance far more severely than uniform drop. Downsampling to 16 beams causes a drop of 36.80% in Moderate Car mAP_{40}.

3) Small and non-rigid categories (Pedestrian, Cyclist) suffer disproportionately under beam reduction. At 16 beams, Pedestrian Moderate mAP_{40} drops to 26.97%, as thin vertical structures lose scanlines required for anchor matching.

3.2 Range-Dependent Breakdown Analysis

Evaluating performance across depth intervals ($Near$, $Medium$, Far) uncovers spatial failure modes obscured by global metrics.

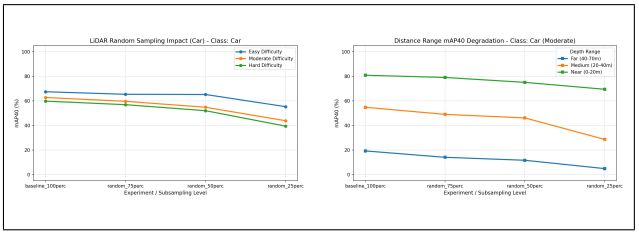


Figure 1: Performance degradation curves across uniform random point drop.

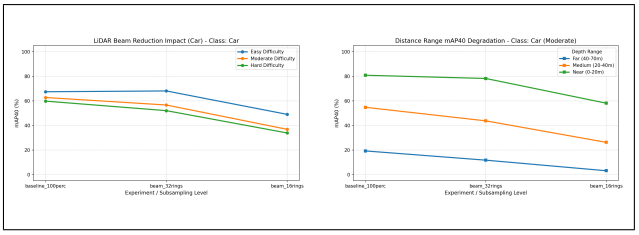


Figure 2: Performance degradation curves across vertical beam reduction.

Spatial Findings:

- **Near Range (0–20 m):** High point density ensures strong resilience (> 55% mAP_{40} retention for Cars at 16 beams).
- **Far Range (40–70 m):** Severe degradation. Under 16-beam sampling or $\rho = 0.25$, distant objects return too low points, falling below the pillar encoder threshold.

4 Conclusion & Future Work

This paper presented a quantitative evaluation of PointPillars under LiDAR sparsification without retraining. Results confirm that while PointPillars is resilient to uniform density drop down to 50%, vertical beam reduction induces severe degradation, especially for far objects (> 40 m) and small categories.

4.1 Future Work

Building upon these findings, our next step is to incorporate RGB camera imagery—already available within the KITTI benchmark—and employ multi-modal sensor fusion algorithms. We aim to investigate whether camera-LiDAR data fusion can effectively compensate for the precision drops and missing geometric cues caused by LiDAR sparsification.